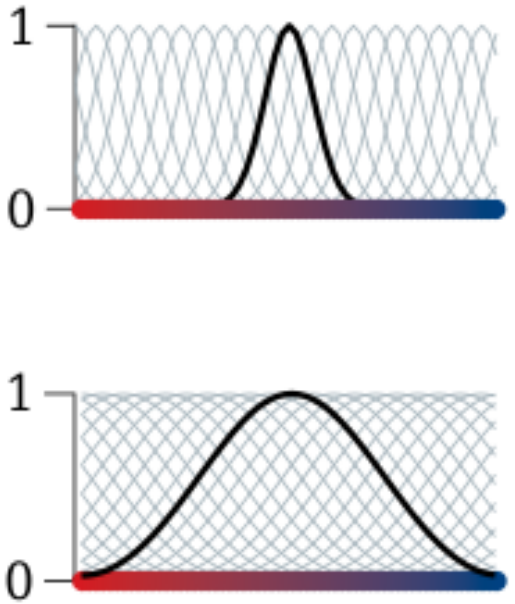
Neural Tuning and Representational Geometry

Kriegeskorte & Wei (2021), Nature Reviews Neuroscience 22: 703-718

Presented by Audrey and Kevin

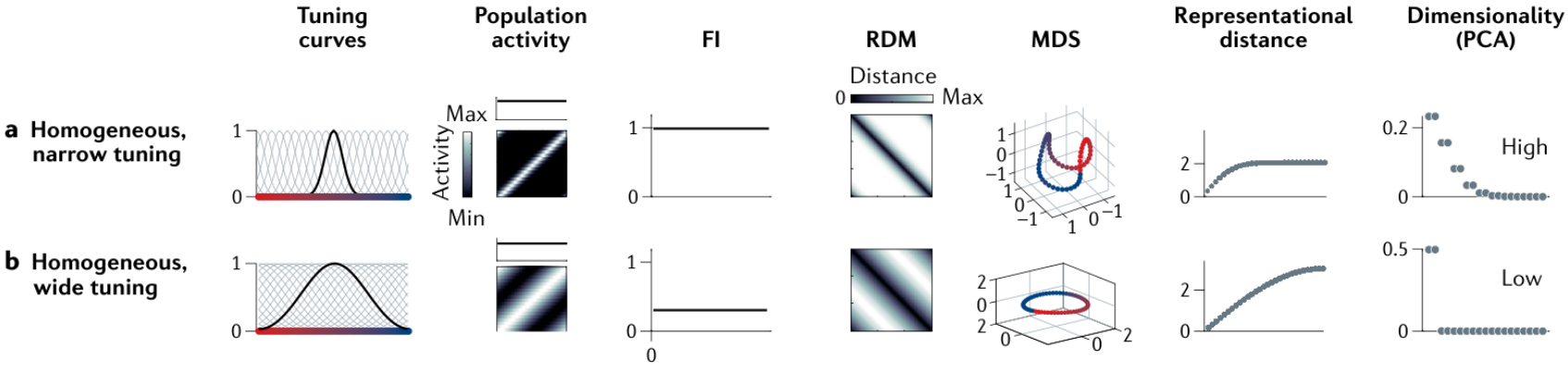
Two Ways to Describe a Neural Code

- Tuning curve: one neuron's average firing rate as a function of a stimulus property (e.g., V1 neurons tuned to edge orientation). Right: one narrowly and one widely tuned neuron (black) among many others (grey).
- Representational geometry: treat the whole population's response to each stimulus as a point, and measure the distances between all pairs of points.
- These distances are stored in a representational dissimilarity matrix (RDM): rows and columns are stimuli, each cell is how different two responses are.
- Goal of the paper: show mathematically how these two views relate.



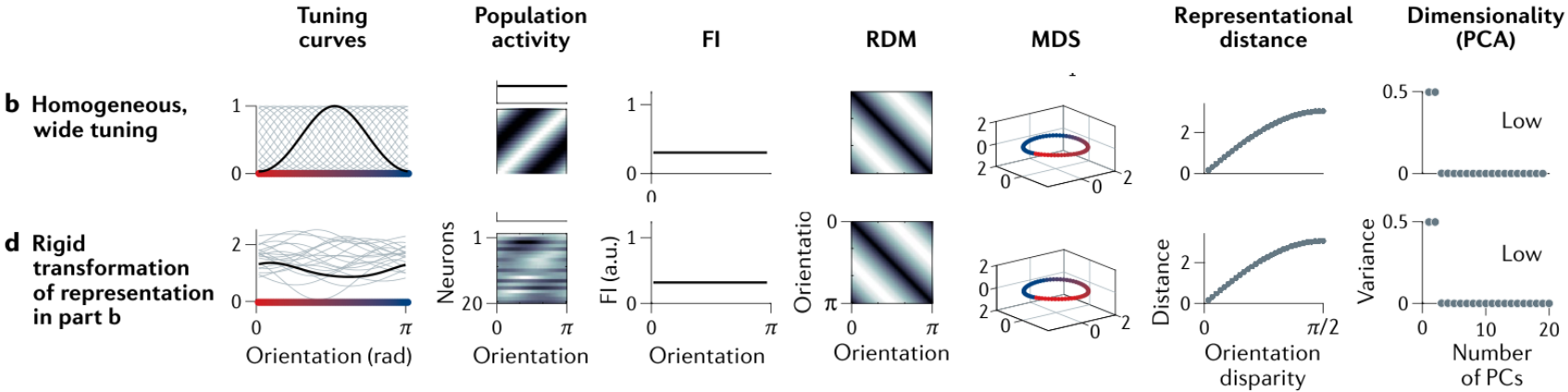
Tuning Determines Geometry (Fig. 1a-b)

- Same population of orientation-tuned neurons, only the tuning width differs.
- Narrow tuning: nearby orientations already look very different, then distance levels off quickly. The patterns spread across many dimensions.
- Wide tuning: distance grows gradually with orientation difference. The patterns form a simple circle in 2 dimensions.
- So changing the tuning changes the geometry, even though both codes cover all orientations.



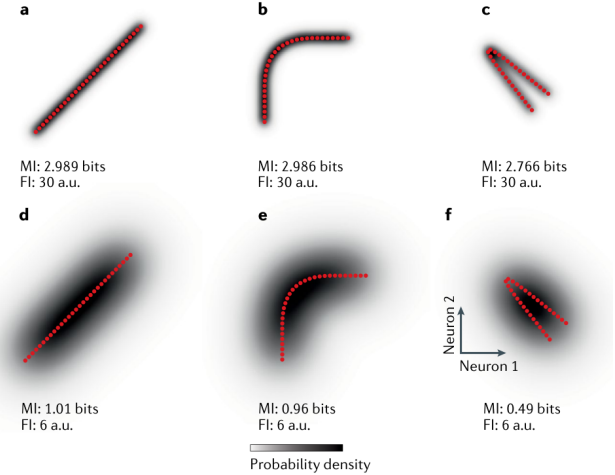
Geometry Does Not Determine Tuning (Fig. 1b vs. 1d)

- Row d is row b rotated and shifted in the space of neural responses.
- The tuning curves look completely different (messy, non-bell-shaped), but the RDM, the MDS picture and the distances are identical.
- So many different sets of neurons can produce the same geometry.
- What the geometry leaves out: whether the information is held by a few selective neurons or spread across many.



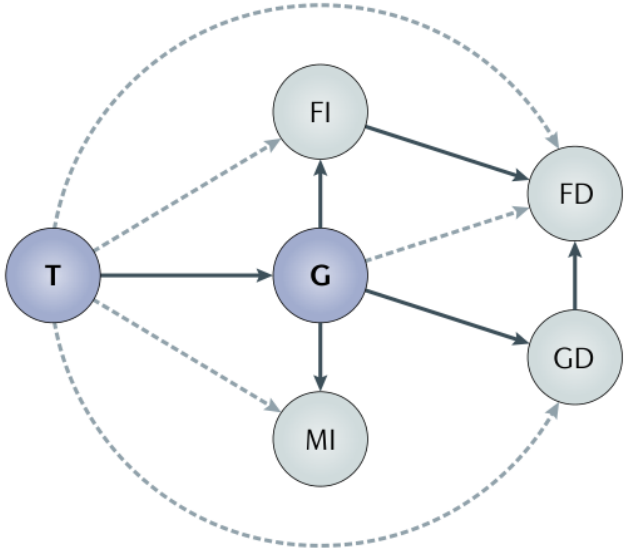
Local Sensitivity Is Not the Whole Story (Fig. 2)

- Fisher information (FI) measures how well the code distinguishes very similar stimuli. Mutual information (MI) measures how much the response tells you about the stimulus overall.
- Each panel: 30 stimuli (red dots) encoded by two neurons. Top row low noise, bottom row high noise.
- Bending or folding the line keeps FI the same (the dots stay equally spaced), but folding lowers MI because opposite ends become confusable.
- Lesson: FI only sees the local geometry; the full geometry is needed to predict what can be told apart.



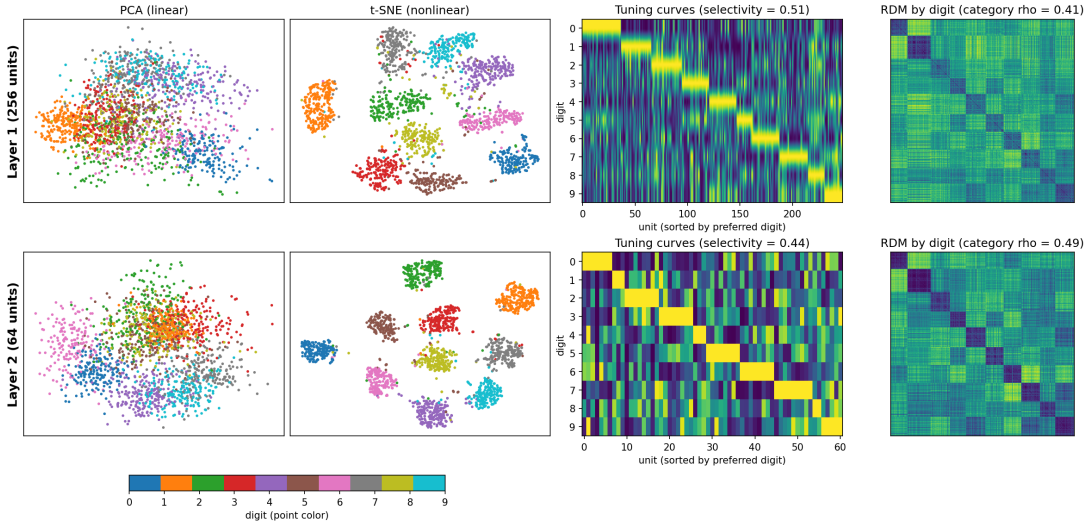
Summary of the Relationships (Fig. 3)

- An arrow means "is enough to determine", assuming we know the noise.
- Tuning (T) determines geometry (G).
- Geometry determines FI, MI, fine discrimination (FD) and general discrimination (GD).
- No arrow goes backward: knowing the geometry does not give you the tuning, and knowing FI does not give you the geometry.
- Also covered in the paper: efficient coding and noise correlations (Fig. 4), which become easier to understand through geometry.



Our Analysis: Two Layers of an MNIST Network

- We trained a small network (784-256-64-10, 96% test accuracy) and compared hidden layer 1 with hidden layer 2.
- Compression: layer 2 needs 8 principal components to explain 90% of the variance, versus 27 in layer 1.
- More digit-like geometry: the RDM matches a "same digit vs. different digit" model better (0.41 to 0.49), and t-SNE clusters are cleaner.
- Less selective tuning: layer 2 units respond to several digits (selectivity 0.51 to 0.44), so digit information is spread across the population.



Takeaways and Discussion

- Tuning describes how individual neurons encode information; geometry describes what the whole population makes available to a reader.
- Tuning determines geometry, and geometry determines FI, MI and discrimination performance, but not the reverse.
- Strength: one clear framework that connects single-neuron studies with population and RSA studies.
- Limitation: theory and toy models only, and it assumes the noise can be made uniform, which may not hold for real neurons.
- Discussion: if geometry is what matters for behaviour, why would the brain prefer one set of tuning curves over another?